Requirements Compliance Matrix

River Hawk Consulting, LLC · IMAX HCD Prototype OT

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Phase 3 Requirements Compliance Matrix

River Hawk Consulting, LLC · UNCLASSIFIED · Re-scored 7 August 2026 Source of requirements: the **executed Other Transaction Agreement No. 5600002690012** (effective 6 August 2026) and its Attachment 1, the Task Description Document — together the governing instrument under Articles XX and XXI. The IMAX HCD Prototype OT solicitation and the Phase 2 selection letter are retained below as the origin of the six deliverables, which the TDD carries forward as D1–D6.

This matrix scores the **solicitation's** ①–⑥. The TDD's own M1–M5 milestones and D1–D6 data deliverables — which are what the Government verifies for acceptance and what Article VIII(b) makes a precondition of invoicing — are scored separately in `milestone_record.md`. A full clause-by-clause audit of the executed agreement, including what it requires that this matrix never covered, is in `signed_ot_audit.md`.

Scored against **`v1.2.1-prototype`**. `v1.1.0-prototype` preceded the 7 August work (twelve review defects, the virtualized message stream the TDD commits to, and the dense Arabic customs document); `v1.0.0-prototype` froze the 1 August build, which River Hawk completed at its own expense **before award**. Where a row depends on newer work it says so.

Legend: ✅ met · 🟡 met with a stated caveat · ⚠️ blocked on Government action

Every claim below was re-checked against the repository — on 4 August against the solicitation, and again on **7 August against the executed agreement**. Claims that were wrong are corrected in place; both checks are recorded in §F.

A. The six required prototype deliverables

> *"The ideal prototype projects provide the following…"*

#	Requirement (as written)	Status	Where it is	Notes
1	**HCD Artifacts** — to be defined by the contractor	✅	`hcd/`, PDF in `project/delivery/`	All four TDD Task 2 subtasks: **2.1** four personas + empathy maps · **2.2** journey map, four task flows, 20 per-persona success criteria · **2.3** persona-indexed annotated wireframes · **2.4** pain-to-design traceability matrix (24 pains, each traced to a design response, screen, specification and screenshot) + a 16-entry assumption log. Plus three workflow-model findings produced during performance. Delivered as one document: `hcd_artifacts.pdf`
2	**UI Codebase** — a single-page application implementing all HCD screens, stored in a GitHub repository	✅	`angular/` on `main`	Angular 21.2.19, pinned exactly (no `^`/`~` on `@angular/core` or `@angular/cli`); standalone components + signals; every wireframed screen implemented; unlimited rights. The 1 August build was frozen at `v1.0.0-prototype`; the SPA changed again on **7 August** to close twelve defects from internal review and to implement the virtualized message stream TDD 3.3 commits to. Every change is covered by a specification and listed in

#	Requirement (as written)	Status	Where it is	Notes
				`signed_ot_audit.md` §3.1
3	**Test Suite** — end-to-end smoke tests	✅	`tests/`	**49 e2e specs** — `npm run test:ng` runs `shell.spec.js` (2) + `services.spec.js` (6) + `tests/ng/` (41) — plus 10 Angular unit specs and 59 Node unit specs covering identity resolution and certificate handling. Includes the TDD Task 5 performance verification: a synthetic 2,000-message thread proves the stream is windowed and measures scroll-to-repaint at ~17 ms against the TDD's 200 ms budget. The evaluated Phase 2 suite is retained and its per-flow status documented
4	**Documentation** — a user guide and a developer guide	✅	`docs/`, PDFs in `project/delivery/`	Both written; a deployment guide (`deploy/README.md`) added beyond the requirement
5	**Project Artifacts** — sprint board export, burndown chart, risk register	✅	`project/acceptance/board_export_final.md`, `project/burndown.md`, `project/risk_register.md`	Maintained daily during performance, not reconstructed at the end
6	**Demo & Acceptance Package** — recorded walkthrough, slide deck summarizing	🟡	`project/acceptance/`	**4:08 narrated walkthrough** captured against the 7 August build and narrated properly — the synthesized scratch track is superseded, and

#	Requirement (as written)	Status	Where it is	Notes
	objectives, test results, and open issues			the whole recording is reproducible (`narration.json` + `scripts/capture-walkthrough.js` + `scripts/build-walkthrough.sh`) rather than a one-off. Deck (+PDF) covering objectives / test results / open issues, evidence trail. **Remaining caveat: the live demonstration M5 also requires is scheduled for 20 August, in person, and has not yet taken place**

B. What the solution was to address

> *"The solution should address…"*

Requirement	Status	Evidence
Converting high-level requirements into **mission personas, HCD wireframes and UI designs**	✅	`hcd/personas.md`, `hcd/empathy_maps.md`, `hcd/journey_map.md`, `hcd/wireframes.md` — wireframe annotations are persona-indexed and cross-referenced to live screenshots; `hcd/traceability_matrix.md` joins each persona pain to the design response, screen, validating specification and screenshot
Leveraging the HCD output to create a **responsive single-page application**	🟡	Fluid flex layout, independently collapsible **and drag-resizable** panels, a resizable/maximizable scan viewer, wrapping action strip, and a queue-only view. **Caveat: no media-query breakpoints** — the build targets the analyst desktop it was evaluated on; small-screen breakpoints were not in scope and are not claimed
Leveraging existing services or providing containerized mock services that emulate production services	✅	Five single-responsibility mock service pods behind frozen JSON contracts, plus a configuration-only path to the Sponsor's live model gateway (`deploy/README.md` §4) — no client rework at cutover
Delivering a **prototype acceptance package** including a live demo of chat-viewer functionality, test scripts, and documentation	✅	`project/acceptance/` + runnable demo (`npm run dev:ng`) + the deployable image set

C. Performance constraints

Constraint	Status	Notes
Period of performance ~**14 calendar days**	✅	**6 – 20 August 2026** per Article II of the executed agreement (Day 1 = 7 August, the first calendar day after execution, per TDD Assumption (1)). River Hawk built the SPA at its own expense ahead of award — complete 1 August — so the period opened with the build already done and was spent on the Platforma environment work, the review defects, and the acceptance package
All work and deliverables unclassified	✅	Entire corpus is fabricated demonstration data; UNCLASS banner rendered by the app shell (`angular/src/app/app.html`) on every screen. All delivered PDFs are footer-marked UNCLASSIFIED
All direct work by US citizens	✅	Asserted by River Hawk Consulting per the Phase 2 OT eligibility submission
Prototype **stored in a GitHub repository**	✅	`Blackfell-Group/imax-chat-viewer`, tagged **`v1.2.1-prototype`** (`v1.0.0-prototype` retained as the pre-award build, `v1.1.0-prototype` as the environment-work freeze). A mirror at `Blackfell-Group/imax-chat-viewer-public` carries **source only** (212 files, squashed history) for the Government-directed transfer route; the deliverable documents travel directly instead (`signed_ot_audit.md` §2.1)
Operates in an **environment compatible with existing IMAX infrastructure** (Platforma)	⚠️	Angular front end per Sponsor direction; six-image containerized set proven on a Kubernetes cluster in CI on every push, and packaged as a Helm chart for ArgoCD (`deploy/chart`) with the kustomize manifests retained and a CI check that the two cannot drift. Identity reads the STS token the front forwards (`AUTH_MODE=bearer-jwt`), and server certificates are read from AWS Secrets Manager at pod start onto tmpfs, so the private key never becomes a Kubernetes Secret. Hardened for restricted transfer (`deploy/AIRGAP.md`): fonts vendored so the SPA fetches nothing at runtime, base images digest-pinned, Dockerfiles parameterized for in-enclave bases and mirrors, registry-prefix transform, optional enclave-CA mount, and an architecture gate that refuses to package images built for the wrong CPU. **Sandbox access was never granted**, so first-run verification inside Platforma remains outstanding, as do the STS claim names and the certificate secret names
Demonstrate end-to-end UI flows	✅	Queue → search → thread → bilingual review → enrichment → OCR → promote to gold → export, proven by the e2e suite and shown in the walkthrough
Unlimited rights in delivered data, including the demonstration recording	✅	Code, documentation, and the recording carry unlimited rights per the Phase 2 IP assertion

D. Beyond the requirement

Built though not required: a deployment guide; a vulnerability-scanning and cluster-deployment CI gate that re-proves the artifact set on every push; a visual-parity baseline against the evaluated demonstration; a running evidence trail; a configuration-only integration path to the Sponsor's model gateway; and an air-gap transfer kit — runbook, CI-built bundle assembled from the scanned images, and a preflight gate that fails the build if the SPA acquires any runtime dependency on the public internet.

Added 3–4 August, after the Sponsor answered the kickoff questions:

- **Identity from the STS token.** The front sends the identity token in `Authorization` and the access token in `x-auth-request-access-token`. The prototype had been reading flat headers, which would have refused every request behind the real front — silently, until the first login. `/api/whoami` reports which claim names a real token carries, so correcting a mismatch is configuration rather than a rebuild.
- **Server certificates from AWS Secrets Manager.** An initContainer reads them at pod start onto an in-memory volume. Both secret layouts and all payload formats the Sponsor might use — raw PEM, JSON, and base64 inside JSON — are detected.
- **A Helm chart for ArgoCD**, proven equivalent to the kustomize manifests by a CI check that fails on any difference, and rolled out on a live cluster rather than only templated.
- **The HCD artifacts as one publication-quality document**, covering all four TDD Task 2 subtasks (§F).

F. The 4 August re-check

Every claim above was re-verified against the repository rather than carried forward from the 1 August scoring. What was checked, and how:

Claim	Method	Result
Angular 21 LTS, pinned	Read `angular/package.json`	`@angular/core` and `@angular/cli` both `21.2.19`, exact — no `^` or `~`
SPA codebase “on `main`”	`git diff main...HEAD -- angular/`	Empty. The claim holds; later work is confined to `deploy/`, `hcd/`, `tests/`, `scripts/`
32 e2e specs	Counted `test()` per file against what `test:ng` actually runs	2 + 6 + 24 = 32. Correct
10 unit specs	Ran `npm test --prefix angular`	10 passed
Project artifacts present	Checked each of the three paths	All present
Narrated demo 2:34	`ffprobe` on the file	2:34
”No media-query breakpoints”	Grepped `angular/src` for layout `@media`	Zero. The 🟡 caveat is honest, not stale
Deck covers objectives, test results, open issues	Read its section headings	All three present
UNCLASS banner on every screen	Located it in the app shell	`angular/src/app/app.html`
Both guides present	Listed `docs/`	`user_guide.md`, `developer_guide.md`

Two things were wrong and are corrected above:

1. **Deliverable ① was over-scored.** It was marked met on the strength of personas, empathy maps and wireframes. Our own TDD commits to four Task 2 subtasks, and two did not exist: **2.2** task flows and journey map with per-persona success criteria, and **2.4** the assumption log and pain-to-design traceability matrix — which the TDD says “is delivered with the artifact package” and which Task 5.2 depends on. Against the solicitation’s “HCD Artifacts — defined by the contractor” the original score was defensible; against the TDD it was not. Both artifacts are now written and the deliverable is rendered as `project/delivery/hcd_artifacts.pdf`.
2. **The scoring was stale.** It was dated 1 August and declared itself scored against tag `v1.0.0-prototype`, while the package had moved five commits past that tag. The header now says what it is actually scored against.

Deliverable ① now maps to the TDD as follows:

TDD Task 2	Artifact
2.1 Proto-personas & empathy maps	`hcd/personas.md`, `hcd/empathy_maps.md`
2.2 Task flows & journey map	`hcd/journey_map.md` — 4 task flows, 20 success criteria, each naming the specification that replays it
2.3 Wireframes	`hcd/wireframes.md`
2.4 Validation & traceability	`hcd/traceability_matrix.md` (24 pains → response → screen → specification → screenshot), `hcd/assumption_log.md` (16 assumptions, 6 closed)

E. Where the build departs from the evaluated demonstration

Three HCD findings changed the product during performance, each an artifact with its rationale and restoration path (`hcd/`): the unit of gold is the thread; no verdict without the source; a linguist's bench with one output. The third **removes** snippet clipping from the evaluated flow. The evaluated React implementation remains in the repository unmodified, and the removal reverts in a single commit if the Government prefers strict parity. Per-flow status of the evaluated acceptance suite is in `project/evidence/README.md`.